	acceleration starts to increase again, in the same direction, until it reaches the same maximum value as at the start at x =28.0 cm.
7(a)(i)	The resistance of a wire is the ratio of the potential difference across it to the current passing through it.
7(a)(ii)	Resistivity ρ of a material is the resistance between two opposite faces of a wire made of the same material with the faces of unit cross-sectional area and of unit length apart and is given by $\rho = \dfrac{RA}{L}$.
7(b)(i)	Using $V = IR$ $6.0 = 1.2(R + 0.1)$ $R = 4.9\ \Omega$
7(b)(ii)	$R = \dfrac{\rho L}{A}$ $L = \dfrac{RA}{\rho} = 81.5$ m number of turns $= \dfrac{L}{2\pi r} = 118$

7(c)	$B = 0.72\mu_o \dfrac{(118)(1.2)}{11\times10^{-2}} = 1.16\text{ mT } = 1.2\text{ mT}$
7(d)(i)	K.E. of electron = eV $\frac{1}{2}mv^2 = eV$ Momentum, $p = mv$. Thus, $\frac{p^2}{2m} = eV$ $p = \sqrt{2meV} = 8.54\times10^{-24}$ N s
7(d)(ii)	For circular motion, net force towards the centre, which is the magnetic force = centripetal force required. $F_B = \frac{mv^2}{r}$ $F_B = Bqv$ Thus, $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq} = \frac{8.54\times10^{-24}}{Bq} = 0.044$ m
7(e)	The component of its velocity parallel to the magnetic field makes the electron moves in the direction of the magnetic field at constant velocity. No magnetic force is present due to the parallel component. The component of the velocity perpendicular to the magnetic field will result in the electron moving in a circular path as the magnetic force present is always directed towards the centre of its path. This circular path has a constant radius. Thus, the electron will trace out a helical path.
8(a)(i)	Radioactive decay is a random process whereby a nucleus becomes more